

# Perception of Hazards: The Role of Social Trust and Knowledge

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Recent research indicates that social trust of those who manage a hazard is strongly correlated to judgments about the hazard's risk and benefits. The present study investigates the more specific question of "For which hazards is this?" It was postulated that when an individual lacks knowledge about a hazard, social trust of authorities managing the hazard determines perceived risks and benefits. On the other hand, when an individual has personal knowledge about a hazard and therefore does not need to rely on managing authorities, social trust is unrelated to judged risks and benefits. Participants ( $N = 91$ ) assessed risks, benefits, and trust in managing authorities and personal knowledge associated with 25 hazardous technologies and activities. As expected, strong correlations between social trust and judged risks and benefits were observed for hazards about which people did not possess much knowledge. No significant correlations between social trust and judged risks and benefits were found for hazards about which people were knowledgeable. Results suggest that the lay public relies on social trust when making judgments of risks and benefits when personal knowledge about a hazard is lacking. Replicating findings of other studies, the present study also found negative correlations between perceived risks and perceived benefits. When social trust was controlled for, correlations between perceived risks and benefits diminished. Implications of the results for risk management are discussed.

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**KEY WORDS:** Risk perception; benefit perception; social trust; knowledge; hazards

## 1. INTRODUCTION

Social science research has demonstrated that technical experts and laypeople often differ in their conclusions about the risks and benefits of hazards (Fischhoff, Slovic, & Lichtenstein, 1982; Sjöberg, 1998). This should be no surprise since they typically also differ in the way that conclusions about risks and benefits are made (Fischhoff *et al.*, 1982). One important difference is that technical experts, because of their training and experience, are more likely to have

knowledge about hazards not available to the average citizen. Lacking this knowledge, lay judgments of risks and benefits may be based on assessments of those who are responsible for managing the hazards.

Recent discussions have recognized the importance of social trust to judgments about risks and benefits and the acceptability of technologies (Cvetkovich & Löfstedt, 1999; Earle & Cvetkovich, 1995; Siegrist, 1999, 2000; Slovic, 1993, 1997). Yet to be answered are questions about for which specific hazards this generality is true. Because technical experts—because of their expertise about particular hazards—are better able to directly assess the risks and benefits of hazards, and because less knowledgeable laypeople are more likely to assess social trust rather than direct hazard risk, the following assumption can be made: Social trust will be significantly related to judg-

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ments of risks and benefits for hazards about which an individual has little knowledge. The present study is designed to test this hypothesis.

### 1.1. Trust and Knowledge

Most people do not possess intricate knowledge of technological risks and benefits. Surveys in the United States and in Europe have shown that a majority of the general population is not able to correctly answer basic questions about gene technology (Durant, Bauer, & Gaskell, 1998) or even general science questions (Miller, 1998). Three out of four adults do not understand basic scientific constructs such as DNA, molecules, or radiation (Miller, 1998). It can be assumed that lacking the knowledge, most people do not directly assess the risks and benefits associated with a technology. Rather, the public has to rely on information provided by experts. Experts, however, are not a homogeneous group; they often differ in their assessments of a technology (Sjöberg, 1998). People cannot evaluate the accuracy or reliability of information about risks and benefits associated with modern technologies given by different sources. In the absence of sufficient knowledge, decisions and judgments are guided by social trust (Earle & Cvetkovich, 1995; Luhmann, 1989). The function of trust is to reduce the complexity people are faced with. In other words, instead of making rational judgments based on knowledge, social trust is employed to select experts who are trustworthy and whose opinions can be believed as being accurate. It has been suggested that people have trust in experts who share the values that they believe are important in a given situation (Earle & Cvetkovich, 1995).

Empirical studies have shown that social trust is an important dimension in attitudes toward technologies. Participants who had social trust in companies as well as in scientists involved in gene technology perceived less risks and more benefits associated with this technology than people not having social trust (Siegrist, 1999, 2000). Within the realm of nuclear or hazardous waste disposal and chemical plant operation, trust had a strong influence on perceptions of the specific technology (Bord & O'Connor, 1992; Flynn, Burns, Mertz, & Slovic, 1992; Groothuis & Miller, 1997; Jungermann, Pfister, & Fischer, 1996). People having trust in the authorities and management responsible for the technology or the plant perceived less risks than people not having trust in those members. Given the low degree of scientific knowledge people possess, these results should not be surprising.

Sjöberg (1999), however, disagrees that trust is an important dimension in risk perception. He reported that trust in politicians in general does not significantly predict perception of overall risk. However, the explanation power of trust depends on how it is operationalized. That measures of social trust in unspecified politicians (e.g., general trust in government in all situations) do not explain perceived risk very well is not surprising, nor is it of much practical or theoretical value.

### 1.2. Perceived Risks and Benefits

Inverse relationships between judged risks and benefits have been observed for different activities and technologies (Alhakami & Slovic, 1994; Frewer, Howard, & Shepherd, 1998). Participants perceiving many benefits associated with a technology assessed less risks (i.e., reduced in severity) than participants viewing the technology as not beneficial. The same relationship was observed when aggregated data were used (Alhakami & Slovic, 1994; Frewer *et al.*, 1998; Gregory & Mendelsohn, 1993). Perceived-risk means were strongly correlated with perceived-benefit means. Technologies viewed as beneficial were associated with less risks than technologies not viewed as beneficial. Based on these findings it has been argued that it may be possible to change perceived risks by changing the perception of benefits (Frewer *et al.*, 1998).

Different explanations for these findings are possible. The need for consistency in beliefs and the tendency to avoid cognitive dissonance may cause the negative association between perceived risks and benefits (Alhakami & Slovic, 1994). In other words, pressure toward consistency results in devaluation of the risks and elevation of the benefits for technologies perceived as good. The opposite effect occurs for technologies viewed as bad.

An alternative explanation for the inverse relationships between perceived risks and perceived benefits has been previously proposed (Siegrist, Cvetkovich, & Roth, 2000). It is hypothesized that social trust simultaneously influences both perceived risks and perceived benefits. For most technologies, the associated risks and benefits are not directly visible; therefore, people rely on risk-benefit information provided by sources they trust. This explanation has strong empirical support. When controlling for the influence of social trust, the correlations between perceived risks and benefits of gene technology vanished or were diminished (Siegrist, 1999, 2000). Similar results have been observed for pesticides, nuclear

power, and artificial sweetener (Siegrist *et al.*, 2000). Trust in authorities, companies, and scientists involved in regulating or using a technology was found to have a positive influence on perceived benefits and a negative influence on perceived risks.

### 1.3. Rationale of the Present Study

The hypothesis that social trust is important in the absence of knowledge has been postulated by different authors (Earle & Cvetkovich, 1995; Luhmann, 1989); however, based on available studies, this assumption has not been empirically tested so far. Lacking the knowledge about risks and benefits associated with a technology, people have to rely on social trust when forced to make judgments. Strong negative correlations between trust and perceived risks—as well as positive correlations between trust and perceived benefits—are expected. On the other hand, when people are familiar with a technology, no correlations between trust in outside authorities and perceived benefits or risks are to be expected. In that case, people rely on their own knowledge in making judgments about risks and benefits, and therefore social trust in experts or authorities is not needed.

The present study attempts to replicate previous studies' findings of the inverse relationships between perceived risks and benefits (Alhakami & Slovic, 1994; Frewer *et al.*, 1998). Previous research has focused on the relationship between perceived risks and benefits; the influence of social trust was not taken into consideration. The present study postulates that controlling for social trust, the associations between perceived risks and benefits diminish. This effect was expected for single technologies or activities as well as for aggregated data.

## 2. METHOD

### 2.1. Participants

Participants in the present analysis included 91 students (30 males and 61 females) enrolled in introductory psychological courses at Western Washington University, who received course credit for their participation. Participants ranged in age from 18 to 28, with a mean age of 19.9 ( $SD = 1.8$ ).

### 2.2. Questionnaire

Participants were asked to make four judgments about 25 activities and technologies. Judgments were to be made on (1) risk of each activity or technology,

(2) perceived benefits of each activity or technology, (3) trust in authorities regulating each activity or technology, and (4) knowledge about the risks and benefits of each activity or technology. The activities and technologies were drawn from a larger set of items used in previous studies (Alhakami & Slovic, 1994). Measurements of perceived risks and perceived benefits were based on questions used by Alhakami and Slovic (1994).

The question on perceived risk asked: "In general, how *risky* do you consider each of the following items to be for the United States society as a whole?" The endpoints of the seven-point scale were labeled "not at all risky" (1) and "very risky" (7).

To assess perceived benefits, the following question was asked: "In general, how *beneficial* do you consider each of the following items to be for the United States society as a whole?" The endpoints of the seven-point scale were labeled "not at all beneficial" (1) and "very beneficial" (7).

The questions "In general, how much *confidence* do you have in the authorities regulating the following items?" and "In general, how much do you *know* about the risks and benefits associated with the following items?" were used to measure trust and self-assessed knowledge, respectively. Scales for these assessments ranged from "no confidence at all" (1) to "high confidence" (7), and "know almost nothing" (1) to "know a lot" (7), respectively.

Each question was printed on the top of the page followed by the list of 25 items. Four versions of the questionnaire were used, in which the order of the four questions was varied. Participants were randomly given one of the four versions of the questionnaire to fill out.

The questions were part of a larger research project. Other questions not related to the present article are not presented here.

## 3. RESULTS

Correlations between perceived risks, perceived benefits, trust in authorities, and self-assessed knowledge for each of the 25 items are shown in Table I. For 15 items, significant negative correlations between perceived risks and perceived benefits were observed. Trust in authorities was significantly correlated with perceived risks for 16 items and with perceived benefits for 19 items, respectively. For most of the items, correlations between self-assessed knowledge and the other variables were not significant.

To test the hypothesis that correlations between perceived risks and perceived benefits are caused by

**Table I.** Correlations Between Perceived Risk, Perceived Benefit, Trust in Authorities, and Self-Assessed Knowledge

| Activity/technology  | PR/PB   | PR/TR   | PB/TR  | PR/SK  | PB/SK   | TR/SK  |
|----------------------|---------|---------|--------|--------|---------|--------|
| Asbestos             | -0.14   | -0.18   | -0.07  | 0.27*  | -0.17   | 0.12   |
| Smoking              | -0.43** | -0.05   | 0.07   | 0.20   | -0.08   | 0.19   |
| Radiation therapy    | -0.31** | -0.26*  | 0.47** | -0.05  | 0.08    | 0.22*  |
| Herbicides           | -0.35** | -0.56** | 0.50** | 0.24*  | 0.17    | -0.07  |
| Microwave ovens      | -0.35** | -0.43** | 0.42** | -0.01  | 0.20    | 0.21*  |
| Automobile travel    | -0.02   | -0.16   | 0.27** | 0.23*  | 0.20    | 0.28** |
| Railroads            | 0.08    | -0.21*  | 0.21*  | 0.20   | 0.29**  | 0.31** |
| Nuclear power        | -0.29** | -0.46** | 0.47** | 0.05   | -0.20   | -0.07  |
| Food preservatives   | -0.27** | -0.45** | 0.38** | 0.18   | 0.04    | 0.11   |
| Motorcycles          | -0.10   | -0.10   | 0.42** | 0.07   | -0.06   | -0.12  |
| DDT                  | -0.33** | -0.24*  | 0.33** | 0.53** | -0.28** | -0.01  |
| Surgery              | -0.04   | -0.33** | 0.18   | -0.15  | 0.10    | 0.26*  |
| Home appliances      | -0.04   | -0.07   | 0.21   | 0.06   | 0.12    | 0.21   |
| Motor vehicles       | 0.06    | -0.11   | 0.42** | -0.10  | 0.23*   | 0.46** |
| Handguns             | -0.36** | -0.44** | 0.32** | 0.07   | 0.07    | 0.02   |
| Alcoholic beverages  | -0.27** | -0.30** | 0.38** | -0.02  | 0.20    | 0.10   |
| Bicycles             | 0.10    | 0.10    | 0.29** | 0.05   | 0.21*   | 0.13   |
| X rays               | 0.02    | -0.17   | 0.16   | -0.11  | -0.05   | 0.20   |
| Pesticides           | -0.33** | -0.25*  | 0.57** | 0.20   | -0.07   | -0.02  |
| Hydroelectric power  | -0.26*  | -0.51** | 0.47** | -0.01  | 0.15    | 0.03   |
| Biotechnology        | -0.30** | -0.51** | 0.47** | 0.01   | 0.17    | -0.05  |
| Chemical fertilizers | -0.43** | -0.37** | 0.57** | 0.22*  | -0.06   | -0.08  |
| Prescription drugs   | -0.12   | -0.31** | 0.33** | 0.08   | -0.01   | 0.07   |
| Fire fighting        | 0.21*   | 0.00    | 0.09   | 0.07   | 0.07    | 0.25*  |
| Commercial aviation  | -0.23*  | -0.39** | 0.52** | -0.10  | 0.28**  | 0.36** |

Note. *N* varies between 84 and 91, reflecting missing values; PR = perceived risk, PB = perceived benefit, TR = Trust in authorities, SK = self-assessed knowledge.

\* $p < 0.05$ .

\*\* $p \leq 0.01$ .

the third variable—trust—partial correlations were computed. Table II shows partial correlation coefficients between perceived risks and perceived benefits, controlling for trust for each item. With the exception of the item “Smoking,” controlling for trust resulted in much smaller associations between perceived risks and perceived benefits.

Means of perceived risks, perceived benefits, trust in authorities, and self-assessed knowledge of the 25 items are presented in Table III. As can be seen in the table, there is a substantial variance among the assessments of the different activities and technologies. Further analyses based on aggregated data were conducted to clarify the differences.

Correlations across the 25 items between perceived-risk means, perceived-benefit means, trust means, and self-assessed knowledge means are shown in Table IV. A strong association between risk and benefit means was observed ( $r = -0.69$ ,  $p < 0.001$ ). Perceived-risk means and trust means were negatively correlated, and perceived-benefit means and trust means were positively correlated. The partial correlation showed that controlling for trust means, the asso-

ciation between perceived-risk means and perceived-benefit means was substantially reduced ( $r = -0.36$ ,  $p < 0.10$ ,  $df = 22$ ). No significant correlation was found between self-assessed knowledge and any other variable. Because of the small sample size, possible outliers may have had a strong influence on Pearson’s correlation coefficients. Therefore, correlations based on the rank information of the data were computed as well. The magnitudes of the coefficients were somewhat different, of course, but the general pattern was found to be the same.

Figure 1 presents a scatter plot of knowledge means and the correlation coefficients for perceived-risk means and trust means. The scatter plot shows that there is a linear relationship between the two variables ( $r = 0.59$ ,  $p < 0.005$ ). The ordinary least squares (OLS) regression line is depicted in Fig. 1. Strong correlations for perceived-risk and trust were observed for the possible risks and benefits of items about which participants did not believe they possessed much knowledge. For activities and technologies associated with a high level of knowledge, trust did not have a negative influence on perceived-risk

**Table II.** Partial Correlations Between Perceived Risk and Perceived Benefit, Controlling for Trust in Authorities

| Activity/technology  | Partial correlation |
|----------------------|---------------------|
| Asbestos             | -0.15               |
| Smoking              | -0.43**             |
| Radiation therapy    | -0.22*              |
| Herbicides           | -0.09               |
| Microwave ovens      | -0.21*              |
| Automobile travel    | 0.03                |
| Railroads            | 0.13                |
| Nuclear power        | -0.09               |
| Food preservatives   | -0.13               |
| Motorcycles          | -0.06               |
| DDT                  | -0.27*              |
| Surgery              | 0.02                |
| Home appliances      | -0.03               |
| Motor vehicles       | 0.12                |
| Handguns             | -0.26*              |
| Alcoholic beverages  | -0.18               |
| Bicycles             | 0.08                |
| X rays               | 0.05                |
| Pesticides           | -0.23*              |
| Hydroelectric power  | -0.02               |
| Biotechnology        | -0.07               |
| Chemical fertilizers | -0.29**             |
| Prescription drugs   | -0.02               |
| Fire fighting        | 0.22*               |
| Commercial aviation  | -0.04               |

Note: *dfs* vary between 81 and 88, reflecting missing values.

\* $p < 0.05$ .

\*\* $p < 0.01$ .

means. The data therefore strongly support the conclusion that trust is an important determinant in judgments of risks in the absence of knowledge.

As shown in Fig. 2, similar results were observed for perceived-benefit means. A marginally significant correlation between the two variables was found ( $r = -0.35$ ,  $p < 0.10$ ). There was, however, an outlier that distorted the linear relationship. Deletion of the item "Asbestos" resulted in a strong association between the variables ( $r = -0.61$ ,  $p < 0.005$ ). Because it may be heavily influenced by an outlier, the OLS regression line would be misleading. Figure 2 depicts a robust regression line that was computed using AXIS (program from the Department of Statistics, The Wharton School, University of Pennsylvania; Stine, 1997). In robust regression, outliers are weighted less than other data points and therefore have less influence on the regression line. The present data support the hypothesis that in the absence of knowledge, social trust influences the assessment of benefits associated with a technology, and that for activities and technologies people are familiar with, trust is not needed for making judgments.

**Table III.** Means of Perceived Risk, Perceived Benefit, Trust in Authorities, and Self-assessed Knowledge Across Subjects

| Activity/technology  | Risk | Benefit | Trust | Knowledge |
|----------------------|------|---------|-------|-----------|
| Asbestos             | 4.09 | 2.05    | 4.07  | 3.40      |
| Smoking              | 6.14 | 1.23    | 3.15  | 6.28      |
| Radiation therapy    | 4.34 | 5.14    | 5.06  | 4.11      |
| Herbicides           | 4.16 | 3.71    | 3.62  | 3.44      |
| Microwave ovens      | 2.78 | 5.15    | 4.41  | 3.91      |
| Automobile travel    | 4.31 | 5.88    | 4.82  | 5.65      |
| Railroads            | 3.28 | 5.53    | 5.07  | 4.11      |
| Nuclear power        | 4.61 | 4.29    | 3.73  | 3.86      |
| Food preservatives   | 3.41 | 4.57    | 3.84  | 3.62      |
| Motorcycles          | 4.60 | 3.58    | 4.43  | 4.24      |
| DDT                  | 4.42 | 2.39    | 3.55  | 3.68      |
| Surgery              | 3.68 | 6.47    | 5.59  | 4.96      |
| Home appliances      | 2.44 | 5.90    | 5.18  | 4.71      |
| Motor vehicles       | 4.36 | 5.89    | 5.11  | 5.51      |
| Handguns             | 6.11 | 2.30    | 2.68  | 5.12      |
| Alcoholic beverages  | 4.99 | 3.03    | 3.48  | 5.86      |
| Bicycles             | 2.50 | 5.63    | 5.42  | 5.57      |
| X rays               | 3.24 | 6.10    | 5.03  | 4.21      |
| Pesticides           | 4.28 | 3.82    | 3.48  | 3.90      |
| Hydroelectric power  | 2.61 | 5.63    | 4.81  | 3.50      |
| Biotechnology        | 3.32 | 5.17    | 4.36  | 2.87      |
| Chemical fertilizers | 4.41 | 3.68    | 3.26  | 3.41      |
| Prescription drugs   | 3.71 | 6.00    | 4.78  | 4.52      |
| Fire fighting        | 4.31 | 6.51    | 5.84  | 4.68      |
| Commercial aviation  | 3.54 | 5.63    | 5.07  | 4.33      |

Note: *N* varies between 84 and 91, reflecting missing values.

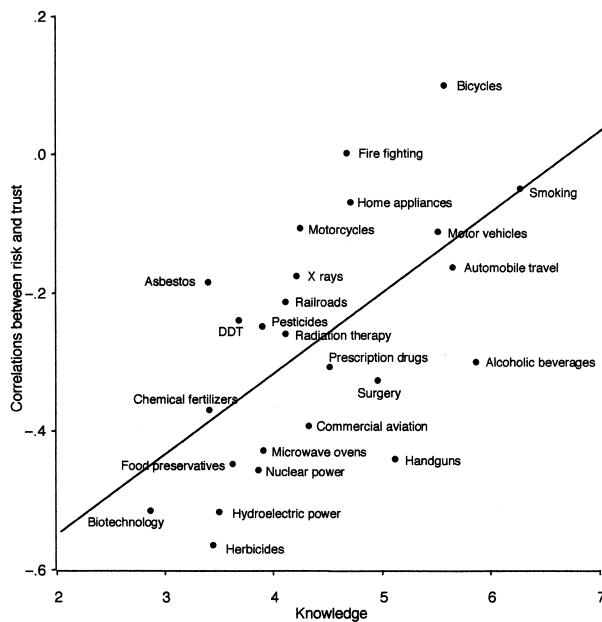
#### 4. DISCUSSION

It has been suggested that one of the functions of trust is to reduce social complexity (Earle & Cveticovich, 1995; Luhmann, 1989). Most laypeople do not possess the detailed knowledge needed for a rational assessment of risks and benefits associated with most complex technologies. They therefore often rely on the opinions of experts or authorities when forced to make risk or hazard assessments. If self-knowledge is perceived as being adequate to assess the risks and benefits of a technology or activity, however, people will make their own judgments. Relying on the opinions of others is not necessary since direct assessments of risks

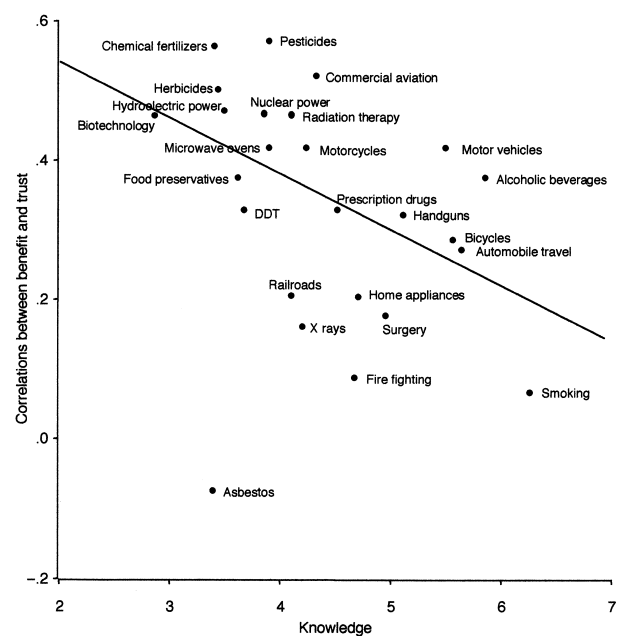
**Table IV.** Correlations Between Perceived Risk, Perceived Benefit, Trust in Authorities, and Self-Assessed Knowledge Based on Aggregated Data ( $N = 25$ )

|           | Risk   | Benefit | Trust | Knowledge |
|-----------|--------|---------|-------|-----------|
| Risk      |        |         |       |           |
| Benefit   | -0.69* |         |       |           |
| Trust     | -0.64* | 0.87*   |       |           |
| Knowledge | 0.37   | -0.00   | 0.12  |           |

\* $p \leq 0.001$ .



**Fig. 1.** Scatter plot of knowledge means and correlations between trust and perceived risks. The OLS (ordinary least squares) regression line is shown.



**Fig. 2.** Scatter plot of knowledge means and correlations between trust and perceived benefits. The robust regression line is shown.

and benefits can be made. The results of the present study strongly support this view of social trust.

It has been argued that the public should be better educated in the area of science so that consumers can form cogent opinions and make rational choices (Frewer, Howard, & Shepherd, 1996). This can be a difficult task, however. People may not be interested. Indeed, many people do not possess the basic scientific knowledge (Miller, 1998) to understand the risks and benefits associated with new technologies. Yet citizens of democratic societies have to reach conclusions and make informed decisions about technological hazards. They may often have to rely on others in making such judgments.

The inverse relationship between perceived risks and perceived benefits found in previous studies (Alhakami & Slovic, 1994; Frewer *et al.*, 1998; Gregory & Mendelsohn, 1993) was successfully replicated in the present study. Significant correlations were observed in 15 of the 25 technologies/activities assessed. People who perceived a technology as risky ascribed less benefits than people perceiving the technology as not risky. The same relationship between the two variables was also found using aggregated data.

It is tempting to assume that perceived risks can be reduced by emphasizing the benefits associated with a technology (Frewer *et al.*, 1998). Results of the

present study suggest, however, that this may not work in a straightforward way. When trust was controlled, substantially reduced correlations between perceived benefits and risks were observed. These findings are in line with other studies in which the influence of trust on gene technology (Siegrist, 1999, 2000), and nuclear power, artificial sweetener, and pesticides (Siegrist *et al.*, 2000) was investigated.

Some methodological issues relevant to the present study have to be addressed. Aggregated data were used to investigate the relationships among knowledge, trust, and perceived risks and benefits. The use of mean values in risk perception research has been criticized as creating a statistical "illusion" (Sjöberg, 1996). Correlations among means are often higher than correlations based on raw data. It cannot be denied that using aggregated data can be somewhat problematic. Not all hypotheses can be tested using raw data, however. In addressing the question of why different technologies are perceived differently, aggregated rather than raw data may be best suited.

The hypothesis that self-knowledge mediates the influence of social trust on perceived risks and benefits was tested across different hazards. The same effect of self-knowledge influencing the association between trust and perceived risks and benefits was expected on an individual level. Participants in the

present study were rather homogenous in their level of knowledge. It was not possible therefore to subdivide the sample population into groups (no knowledge/much knowledge) and to compute separate correlations for each. Future research should focus on assessments of experts compared to laypeople. It is likely that no correlations between social trust and perceived risks or benefits would be observed for those having extensive technological knowledge.

It could be argued that results of the present study are limited because a "convenience" sample (predominantly of female students) was used. Comparing the present results with those of Alhakami and Slovic (1994)—in which a representative Canadian sample was used—suggests that this is not a serious limitation. The observed correlations between perceived risks and perceived benefits in the two studies are similar. There are no plausible reasons to suggest that students should differ from other laypeople in respect to the psychological processes studied. Using a convenience sample for testing basic psychological mechanisms is, therefore, a common practice.

The design used in the present study to investigate risk perception of different technologies has a long tradition (Boholm, 1998; Slovic, 1987). Participants are asked to evaluate a list of items on a number of dimensions. The advantage of this approach is that perceptions of different technologies can be compared with each other. Of course, the same dimensions must be used to evaluate all items. In the present study, social trust was assessed using a single item: Participants were asked how much confidence they had in authorities regulating each specific activity and technology. In complex societies, trust in other institutions and organizations (e.g., companies) that regulate or produce a technology is also an important decision-making component. The source to trust, however, will depend on the relevant technology or activity. It may be that for some aspects of human endeavor, for example, how heavily armed a country should be, trust in one's government will be important. On the other hand, trust in scientific experts may be more important for lay perceptions of risks and benefits when assessing genetically modified foods.

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